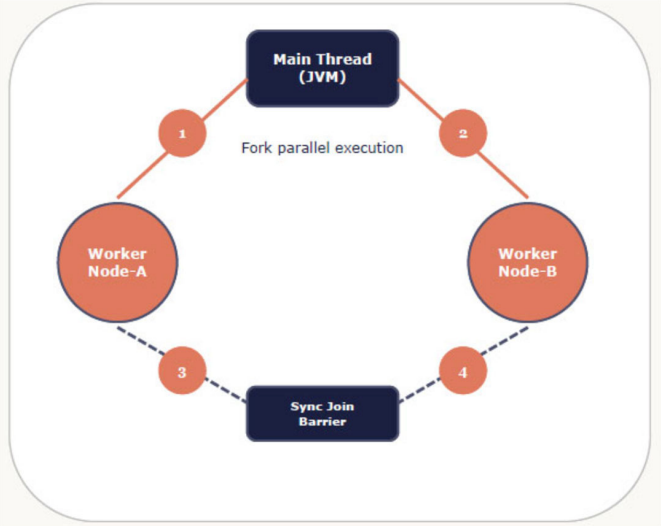
Answer all the questions from 1 to 4 and question number 5 is optional

1- **Compare** between the following:
A-process and thread.
B- process-based multitask and thread-based multitask

2- According to thread in java, there are **three different mechanisms for creating threads**, write these mechanisms with illustrating using code of java to show how to create each one of them.

3- According to the following figure, write a simple program in java for simulating **two independent nodes (Node-A and Node-B)** in a distributed system executing tasks **concurrently,** to perform their work at the same time and ensure that the main program waits until both nodes finish their task.



4- Write a program in java to simulate **multiple clients on one machine (**multiple client threads.**)** and have them **send requests simultaneously** to a server in Java. Consider the following: - each thread acts as an independent client with its own socket connection. - Create 3 client threads. - Each thread connects to the server on port 9000. - Each client sends: "I'm Client X, send me a message when u ready "

5- According to question number 4, write a program in java to detect the maximum number of requests to make the server not overloaded.

I hope you answer all the questions with my best wishes